

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: INTERNET PROGRAMMING
COURSE CODE: CSA43

COURSE OUTCOME COVERED

CO 4: Formulate data-driven web solutions using XML, XSLT, and JSP within the MVC framework. (L4)

Assessment Tool : Class Test

Weightage: 10%

QUESTIONS:

Total Marks: 20

1. Explain the fundamental concepts used in web development by defining the following terms with suitable explanations: **XML, JSP, MVC architecture, XSLT, XML Schema, JSTL, PHP, Database Connectivity, and Parser**. Your answer should clearly describe the purpose and role of each concept in web applications.
2. Explain how the above technologies are interconnected in building a web application. Describe how XML is structured and validated, how JSP and PHP are used for server-side processing, how MVC architecture organizes the application, and how database connectivity and parsers contribute to data processing and communication.

Code of Conduct:

I VARSHA S 192462003 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

RUBRICS FOR CALCULATION:

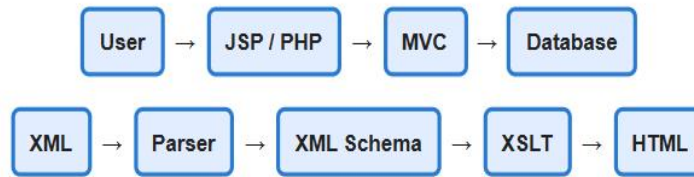
Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Concept Understanding	25%	Clearly explains all concepts with complete accuracy and depth	Minor gaps in understanding	Basic understanding with some errors	Incorrect or very limited understanding
Content Coverage	25%	Covers all required topics comprehensively	Covers most topics with minor omissions	Covers some topics only	Very few topics covered
Technical Accuracy	20%	All explanations are technically correct and precise	Minor technical errors	Some incorrect or vague explanations	Major technical errors
Integration of Concepts	15%	Effectively connects and relates concepts logically	Some connections made with minor gaps	Limited connections between concepts	No clear relationship between concepts
Clarity & Presentation	15%	Well-structured, clear, and easy to understand	Mostly clear with minor issues	Somewhat unclear or poorly organized	Poorly structured and difficult to understand

1. Fundamental Concepts

Concept	Short Explanation
XML	Markup language used to store and exchange structured data.
JSP	Java server-side technology used to create dynamic web pages.
MVC	Architecture that separates Model, View and Controller for better organization.
XSLT	Transforms XML data into HTML, XML or text.
XML Schema	Defines and validates the structure and data types of XML.
JSTL	Provides standard tags for loops, conditions and formatting in JSP.
PHP	Server-side scripting language used to create dynamic websites.
Database Connectivity	Allows applications to store, retrieve, update and delete database data.
Parser	Reads and processes structured data such as XML.

2. Interconnection of Technologies

These technologies work together to process, validate, store and display web application data.



How they work together

- **XML:** Stores structured data.
- **XML Schema:** Validates XML data.
- **Parser:** Reads and processes XML.
- **XSLT:** Converts XML into HTML.
- **JSP/PHP:** Processes requests and generates dynamic pages.
- **MVC:** Separates data, logic and presentation.
- **Database Connectivity:** Connects the application to the database.
- **JSTL:** Simplifies logic inside JSP pages.

Simple Example

XML → Parser → Validation → XSLT → HTML User → JSP/PHP → MVC → Database → Response

RESULT

The fundamental web development concepts such as **XML, JSP, MVC, XSLT, XML Schema, JSTL, PHP, Parser, and Database Connectivity** were successfully studied. Their roles and interconnection in developing dynamic and structured web applications were understood.